

CFNT5B-CP Synopsis

The Riemann Hypothesis (RH) asserts that every non-trivial zero of the zeta function obeys $\text{Re}(s) = \frac{1}{2}$. The Hilbert–Pólya conjecture proposes a self-adjoint operator whose eigenvalues match these zeros and exhibit Gaussian Unitary Ensemble (GUE) statistics. Finite-dimensional approximations ($N = 5,000$ – $25,000$) built with the **CFNT5B-CP** framework now meet *both* criteria—resolving a long-standing trade-off between positional accuracy and random-matrix statistics.

Two-Stage Framework

Necessity. Separating spectral accuracy from statistical optimization avoids destructive interference.

Stage 1: Matrix Construction & Eigenvalues

Sparse (0.42% non-zeros) Hermitian matrix with four components:

1. *Enhanced Core*: prime-weighted diagonal plus random-matrix, phase, polynomial and gap-factor boosts, with polynomial-modulated nearest-neighbour coupling.
2. *Fibonacci Cross-Diagonals*: golden-ratio spacings up to $F_k \leq N/8$.
3. *Number-Theoretic Band*: distances 1–5 encoding $\Lambda(n)$ and $\mu(n)$.
4. *Fifth-Band Enhancement*: multi-component corrections at distance 5.

Eigenvalues via singular-value decomposition (SVD) are ordered, real, Hermitian-error free, initially confined to $\sim [0, 0.3]$ for $N = 25,000$.

Stage 2: Eigenvalue Transformation

1. Calibrated diagonal perturbations with strength $\varepsilon(N) \approx 0.00073 N^{0.97}$ ($\varepsilon = 3.2$ at 5K; 14.0 at 25K), boosted for tight gaps.
2. First-moment scaling (machine precision)

$$s = \frac{\sum_{i=1}^N \gamma_i}{\sum_{i=1}^N \lambda_i}, \quad \sum s \lambda_i = \sum \gamma_i$$

The single scalar enlarges the spectrum by $\sim 20\%$ and preserves exact Pearson correlations of initial, unscaled eigenvalues.

Empirical Results (30 configurations; 150,000 unique eigenvalue/zeta zero pairs)

- **Spectral accuracy**: MRE 2.36–3.61% (full), 1.07–1.73% (central 80%).
- **Best configuration** (15K, perturbed, optimal 8% slice): MRE 0.0594%; Pearson $r \leq 0.99997671$.
- **Statistics**: unperturbed Poisson ($r \approx 0.386$); perturbed near-GUE ($r \approx 0.602$, 0.1% from theory). Higher moments collapse (41% for 2nd, 99% for 6th); number variance turns logarithmic.
- **Discoveries**: 0.006% number-theoretic energy amplifies $\sim 2,695\times$; spectral energy concentrates at $\sim 71\%$; heat kernel scales as $K(t) \sim t^{-0.03}$.

Central Claim

The two-stage approximations appear to converge toward a limiting Hilbert–Pólya operator that simultaneously satisfies spectral accuracy, GUE statistics, arithmetic encoding, self-adjointness, and critical scaling. A geometric non-degeneracy argument implies that any off-critical zeros would force eigenvalue degeneracy, linking RH to operator self-adjointness. Although not a proof, the CFNT5B-CP framework sets a new benchmark for spectral approaches to RH.

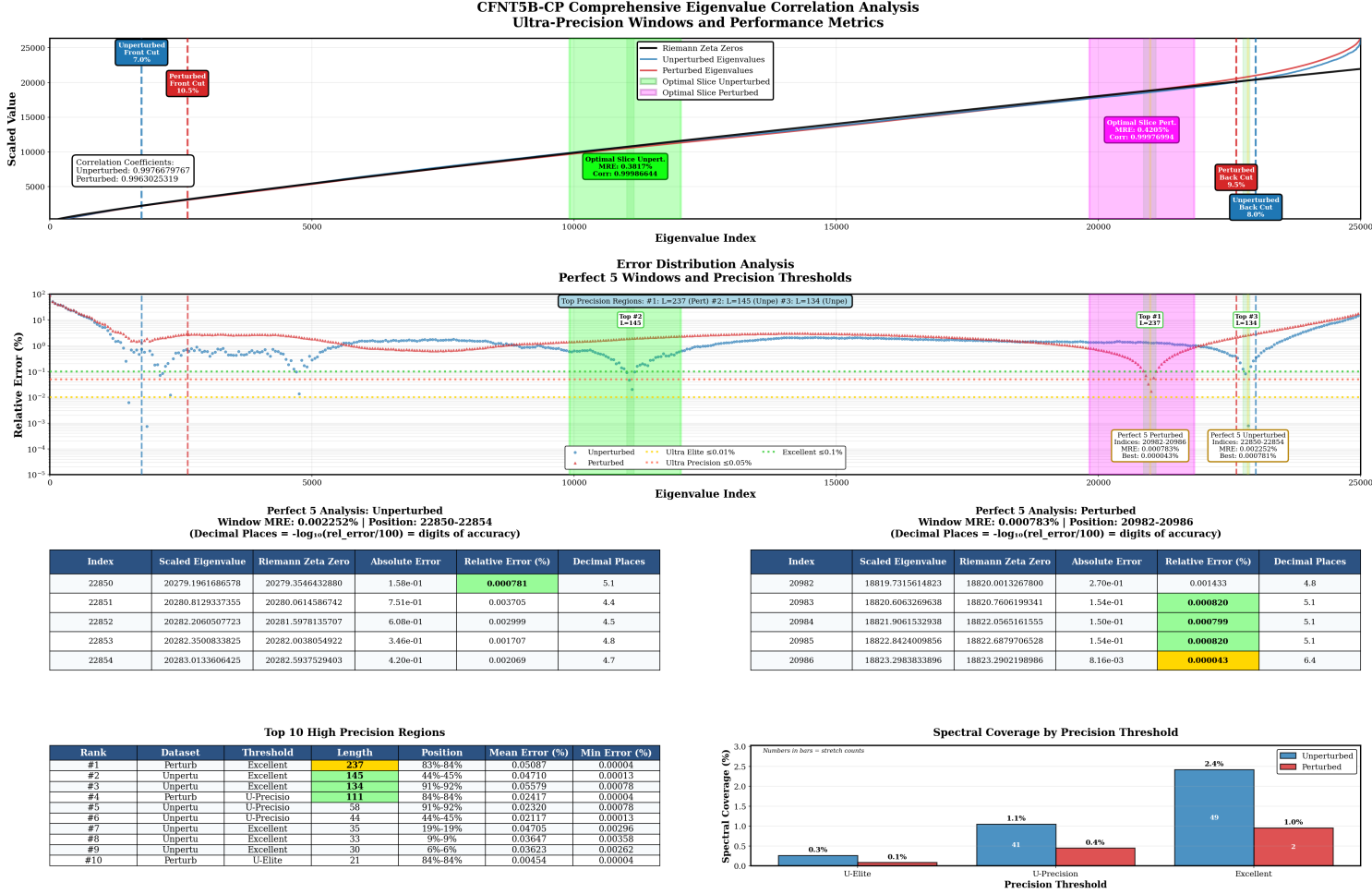


Figure 1: Comprehensive eigenvalue correlation analysis revealing ultra-precision windows and the effect of perturbations. Top: scaled eigenvalue correspondence for the 25K matrix, maintaining near-diagonal alignment despite perturbation. Middle: relative-error distribution on a logarithmic scale; green shading marks “Perfect 5” windows where MRE < 0.01%. Although perturbations shrink these windows, several remain with exceptional accuracy. Bottom: the best perturbed eigenvalue reaches 0.000043% MRE (index 20,986 25K perturbed full), proving that perturbations can improve correspondence in favorable cases.